

SEVERE CHEMICAL & PRECISION MEDICAL THERMAL MANAGEMENT

Specialized Application Process Chillers

Custom Metallurgy & High-Reliability Redundant Refrigeration Systems for Severe Duty Cycles

Engineered Beyond Standard Water Chilling: Specialized industrial processes involve aggressive electrolytes, severe exothermic chemical kinetics, or zero-downtime medical diagnostic imaging that quickly destroy conventional chillers.

Win Equipments designs custom chillers with Grade 2 Pure Titanium immersion evaporators, Hastelloy C-276 and Teflon (PTFE) thermo-coils, sealless magnetic drive chemical pumps, dual-circuit N+1 standby compressors, and automated emergency city water bypass systems.



GRADE 2 TITANIUM

N+1 REDUNDANT

HASTELLOY & TEFLON

CITY WATER BYPASS

THREE SPECIALIZED ENGINEERING PILLARS

1. WAN Anodizing Chillers

Grade 2 Titanium immersion coils impervious to 20% H₂SO₄ baths. Sized for rectifier thermal loads. Type II (+18°C) and Type III Hard Coat (-2°C) operation.

2. WMS Medical Scan Chillers

MRI, CT & LINAC cooling with dual N+1 refrigeration loops, motorized emergency city water bypass to prevent helium quench, and quiet <65 dBA hospital design.

3. WAC Acid Cooling Chillers

Titanium, Hastelloy C-276, and PTFE exchangers with sealless PVDF magnetic pumps for sulfuric pickling, exothermic reactor jackets, and battery formation baths.

TARGET SECTORS

Architectural Aluminum Anodizing

Hospital Diagnostic Imaging (GE, Siemens, Philips)

Lead-Acid & Lithium Battery Formation

Steel Wire & Tube Pickling Halls

Exothermic Chemical Reactor Vessels

ELECTROCHEMICAL SURFACE FINISHING • WAN SERIES

WAN Series Titanium Anodizing Chillers (2 to 100+ TR)

Rectifier Thermal Sizing Guide: Tonnage (TR) = (Rectifier Volts × Amperes × 3.412) / 12,000 × 1.15

MODEL NO.	CAPACITY	COOLING	RECTIFIER CURRENT	EVAPORATOR METALLURGY	COMPRESSOR	ACID FLOW	REFRIGERANT
Model Code	TR	kW	Amps @ 15–18V	Wetted Material	Type	LPM	Gas
WAN 020	2.0 TR	7.0 kW	Up to 350 A	Pure Titanium Gr. 2 Coil/PHE	Copeland Scroll	25 LPM	R410A / R407C
WAN 030	3.0 TR	10.5 kW	Up to 600 A	Pure Titanium Gr. 2 Coil/PHE	Copeland Scroll	35 LPM	R410A / R407C
WAN 050	5.0 TR	17.5 kW	Up to 1,000 A	Pure Titanium Gr. 2 Coil/PHE	Copeland Scroll	55 LPM	R410A / R407C
WAN 075	7.5 TR	26.3 kW	Up to 1,500 A	Pure Titanium Gr. 2 Coil/PHE	Copeland / Danfoss	80 LPM	R410A / R407C
WAN 100	10.0 TR	35.1 kW	Up to 2,000 A	Pure Titanium Gr. 2 Coil/PHE	Copeland / Danfoss	110 LPM	R410A / R407C
WAN 150	15.0 TR	52.7 kW	Up to 3,000 A	Pure Titanium Gr. 2 Coil/PHE	Dual Copeland Scroll	160 LPM	R410A / R407C
WAN 200	20.0 TR	70.3 kW	Up to 4,000 A	Pure Titanium Gr. 2 Coil/PHE	Dual Copeland / Danfoss	215 LPM	R410A / R407C
WAN 300	30.0 TR	105.5 kW	Up to 6,000 A	Pure Titanium Shell & Tube	Bitzer Semi-Hermetic	320 LPM	R407C / R134a
WAN 500	50.0 TR	175.8 kW	Up to 10,000 A	Pure Titanium Shell & Tube	Bitzer / Frascold Semi-Herm	530 LPM	R407C / R134a
WAN 1000	100.0 TR	351.6 kW	Up to 20,000 A	Pure Titanium Shell & Tube	Twin Semi-Hermetic / Screw	1,050 LPM	R407C / R134a

Type II Conventional vs Type III Hard Coat Operation: WAN chillers maintain conventional baths at +18°C to +21°C for optimal dye pore uniformity. For Type III Hard Coat anodizing, chillers operate at -2°C to +4°C with propylene glycol brine and electronic expansion valves.

CLINICAL DIAGNOSTICS & RADIOTHERAPY • WMS SERIES

WMS Series Medical Scan Chillers (3 to 30 TR)

Redundant MRI Superconducting Magnet, CT Tube & LINAC Radiotherapy Head Cooling

MODEL NO.	CAPACITY	COOLING	COMPRESSORS	PUMPS	FLOW RATING	BUFFER TANK	BYPASS VALVE	NOISE
Model Code	TR	kW	N+1 Dual Setup	Dual SS316	LPM @ bar	SS304	Motorized	dBA
WMS 030	3.0 TR	10.5 kW	2 × 1.5 TR Hermetic Scroll	2 × 0.75 HP SS316	32 LPM @ 3.5 bar	80 L	1" Motorized	< 62 dBA
WMS 050	5.0 TR	17.5 kW	2 × 2.5 TR Hermetic Scroll	2 × 1.5 HP SS316	52 LPM @ 3.8 bar	120 L	1.25" Motorized	< 63 dBA
WMS 075	7.5 TR	26.3 kW	2 × 3.8 TR Hermetic Scroll	2 × 2.0 HP SS316	78 LPM @ 4.0 bar	180 L	1.5" Motorized	< 64 dBA
WMS 100	10.0 TR	35.1 kW	2 × 5.0 TR Hermetic Scroll	2 × 3.0 HP SS316	105 LPM @ 4.2 bar	250 L	1.5" Motorized	< 65 dBA
WMS 150	15.0 TR	52.7 kW	2 × 7.5 TR Hermetic Scroll	2 × 4.0 HP SS316	160 LPM @ 4.5 bar	350 L	2" Motorized	< 66 dBA
WMS 200	20.0 TR	70.3 kW	2 × 10.0 TR Hermetic Scroll	2 × 5.5 HP SS316	210 LPM @ 4.5 bar	500 L	2.5" Motorized	< 67 dBA
WMS 300	30.0 TR	105.5 kW	2 × 15.0 TR Hermetic Scroll	2 × 7.5 HP SS316	315 LPM @ 4.8 bar	750 L	3" Motorized	< 68 dBA

Helium Quench Protection Architecture: Integrated motorized three-way bypass valve automatically switches to treated city water in the event of total facility power loss, preserving liquid helium cold heads and preventing catastrophic ₹30L+ helium boil-off.

SEVERE CHEMICAL PROCESSING • WAC SERIES
WAC Series Acid Cooling Process Chillers (2 to 150 TR)

Corrosion-Proof Titanium, Hastelloy C-276 & Teflon (PTFE) Metallurgy for Pickling & Battery Formation

MODEL NO.	CAPACITY	COOLING	EXCHANGER METALLURGY	COMPRESSOR	ACID PUMP	FLOW RATE	GAS
Model Code	TR	kW	Wetted Metal	Type	Sealless Magnetic	LPM	Gas
WAC 020	2.0 TR	7.0 kW	Titanium Gr. 2 / Hastelloy PHE	Copeland Scroll	PVDF Magnetic Drive	25 LPM	R410A / R407C
WAC 030	3.0 TR	10.5 kW	Titanium Gr. 2 / Hastelloy PHE	Copeland Scroll	PVDF Magnetic Drive	35 LPM	R410A / R407C
WAC 050	5.0 TR	17.5 kW	Titanium Gr. 2 / Hastelloy PHE	Copeland Scroll	PVDF Magnetic Drive	55 LPM	R410A / R407C
WAC 075	7.5 TR	26.3 kW	Titanium Gr. 2 / Hastelloy PHE	Copeland / Danfoss	PVDF Magnetic Drive	80 LPM	R410A / R407C
WAC 100	10.0 TR	35.1 kW	Titanium Gr. 2 / Hastelloy PHE	Copeland / Danfoss	PVDF Magnetic Drive	110 LPM	R410A / R407C
WAC 150	15.0 TR	52.7 kW	Titanium / Hastelloy PHE	Dual Copeland Scroll	PVDF Magnetic Drive	165 LPM	R410A / R407C
WAC 200	20.0 TR	70.3 kW	Titanium / Hastelloy PHE	Dual Copeland Scroll	PVDF Magnetic Drive	220 LPM	R410A / R407C
WAC 300	30.0 TR	105.5 kW	Titanium Shell & Tube / PHE	Bitzer Semi-Hermetic	PVDF Magnetic Drive	330 LPM	R407C / R134a
WAC 500	50.0 TR	175.8 kW	Titanium Shell & Tube / PHE	Bitzer Semi-Hermetic	Heavy PVDF Magnetic	550 LPM	R407C / R134a
WAC 1000	100.0 TR	351.6 kW	Twin Titanium Shell & Tube	Dual Bitzer / Screw	Custom Pumping Skid	1,100 LPM	R407C / R134a
WAC 1500	150.0 TR	527.4 kW	Twin Titanium Shell & Tube	Bitzer / Hanbell Screw	Custom Pumping Skid	1,650 LPM	R407C / R134a

Request Technical Submittal & Commercial Quotation

Direct factory engineering consultation for custom capacity sizing, P&ID schematics & dimensional drawings.

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